
SECOND-ORDER LOW-RANK COMPRESSION INSPIRED BY OBS

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1 Feasibility Re-evaluation and Revised Method

1.1 Honest Feasibility Assessment

Let $W \in \mathbb{R}^{m \times n}$ denote one linear layer. The fully general second-order rank-constrained problem is

$$\min_{\text{rank}(\widehat{W}) \leq k} \frac{1}{2} \text{vec}(\widehat{W} - W)^\top H \text{vec}(\widehat{W} - W), \quad (1)$$

where $H \in \mathbb{R}^{mn \times mn}$ is the layer Hessian block.

The exact dense version of (1) is not practical for LLM-scale layers:

1. storing H requires $O((mn)^2)$ memory;
2. a full tangent-space correction over all discarded directions introduces

$$k^2 + k(m - k) + k(n - k)$$

degrees of freedom, which is too large when m, n, k are large.

Therefore, a feasible second-order method should satisfy the following constraints:

1. use a structured Hessian approximation;
2. update singular vectors only in a *small buffered subspace*;
3. solve only small linear systems or low-dimensional least-squares problems;
4. preserve factorized inference form.

Based on these constraints, we propose a revised method with three components:

1. bi-whitened generalized SVD under a Kronecker curvature model;
2. global rank allocation by predicted second-order loss;
3. buffered Gauss–Newton correction in a local low-rank manifold.

1.2 Calibration Statistics and Kronecker Curvature Model

Let the calibration inputs for a layer be

$$X = [x_1, \dots, x_N] \in \mathbb{R}^{n \times N},$$

and let

$$G = [g_1, \dots, g_N] \in \mathbb{R}^{m \times N}$$

collect the output gradients $g_t = \partial \mathcal{L} / \partial y_t$ on the same calibration samples. Define

$$A := \frac{1}{N} X X^\top + \varepsilon_A I_n, \quad B := \frac{1}{N} G G^\top + \varepsilon_B I_m, \quad (2)$$

where $\varepsilon_A, \varepsilon_B > 0$ ensure numerical stability.

We approximate the Hessian block by the Kronecker form

$$H \approx A \otimes B. \quad (3)$$

This yields the weighted local objective

$$\min_{\text{rank}(\widehat{W}) \leq k} \frac{1}{2} \|B^{1/2}(W - \widehat{W})X\|_F^2. \quad (4)$$

Since $A = \frac{1}{N}XX^\top + \varepsilon_A I$, (4) is equivalent, up to the factor N , to

$$\min_{\text{rank}(\widehat{W}) \leq k} \frac{N}{2} \text{vec}(\widehat{W} - W)^\top (A \otimes B) \text{vec}(\widehat{W} - W). \quad (5)$$

Remark. If output gradients are unavailable, one may set $B = I_m$. In that case, the method reduces to a one-sided whitening formulation.

1.3 Exact Base Solution: Bi-Whitened Generalized SVD

Let

$$M := B^{1/2}WA^{1/2}. \quad (6)$$

For any candidate $\widehat{W}$, define

$$Y := B^{1/2}\widehat{W}A^{1/2}. \quad (7)$$

Then $\text{rank}(Y) = \text{rank}(\widehat{W})$, and (5) becomes

$$\min_{\text{rank}(Y) \leq k} \frac{N}{2} \|Y - M\|_F^2. \quad (8)$$

Let the singular value decomposition of M be

$$M = \widetilde{U}\widetilde{\Sigma}\widetilde{V}^\top, \quad \widetilde{\Sigma} = \text{diag}(\widetilde{\sigma}_1, \dots, \widetilde{\sigma}_r), \quad (9)$$

with $\widetilde{\sigma}_1 \geq \widetilde{\sigma}_2 \geq \dots \geq \widetilde{\sigma}_r \geq 0$.

By Eckart–Young, the exact rank- k minimizer of (8) is

$$Y_k^* = \widetilde{U}_k \widetilde{\Sigma}_k \widetilde{V}_k^\top, \quad (10)$$

and the corresponding weight-space solution is

$$\boxed{\widehat{W}_k^{(0)} = B^{-1/2} \widetilde{U}_k \widetilde{\Sigma}_k \widetilde{V}_k^\top A^{-1/2}.} \quad (11)$$

The minimum predicted layer loss is

$$\boxed{\Delta \mathcal{L}_k^{\text{pred}} = \frac{N}{2} \sum_{i>k} \widetilde{\sigma}_i^2.} \quad (12)$$

Interpretation. Equation (11) updates:

1. the singular values;
2. the left singular vectors;
3. the right singular vectors.

Thus, unlike fixed-basis methods, both singular subspaces move.

1.4 Global Rank Allocation by Predicted Second-Order Loss

Suppose layer ℓ has dimensions $m_\ell \times n_\ell$, generalized singular values

$$\tilde{\sigma}_{\ell,1} \geq \tilde{\sigma}_{\ell,2} \geq \dots \geq 0,$$

and chosen rank k_ℓ . Storing the low-rank factors requires

$$P_\ell(k_\ell) = (m_\ell + n_\ell)k_\ell \quad (13)$$

parameters (ignoring bias terms). Under the base solution (11), the predicted compression loss of layer ℓ is

$$\Delta_\ell(k_\ell) = \frac{N}{2} \sum_{i>k_\ell} \tilde{\sigma}_{\ell,i}^2. \quad (14)$$

Therefore the global rank allocation problem is

$$\min_{\{k_\ell\}} \sum_{\ell=1}^L \Delta_\ell(k_\ell) \quad \text{s.t.} \quad \sum_{\ell=1}^L (m_\ell + n_\ell)k_\ell \leq P_{\text{budget}}. \quad (15)$$

Dropping the i -th retained generalized singular component in layer ℓ increases the predicted loss by

$$\frac{N}{2} \tilde{\sigma}_{\ell,i}^2$$

and saves

$$m_\ell + n_\ell$$

parameters. Hence the marginal loss-per-parameter score is

$$\rho_{\ell,i} = \frac{\tilde{\sigma}_{\ell,i}^2}{m_\ell + n_\ell}. \quad (16)$$

A greedy global allocator therefore drops components with the smallest $\rho_{\ell,i}$ first until the budget is satisfied.

1.5 Why a Further Update is Still Useful

Under the exact Kronecker model $H = A \otimes B$, the base solution (11) is already optimal for the local quadratic objective. However, in practice, three sources of mismatch remain:

1. H is only approximately Kronecker;
2. A and B are estimated from finite calibration data;
3. earlier compressed layers perturb the input distribution seen by later layers.

Therefore, a small correction after the base generalized SVD can still improve the actual calibration error.

1.6 Buffered Gauss–Newton Correction

Let the top k generalized singular triplets of M be kept, and let the next p discarded triplets form a *buffer*. Partition

$$\tilde{U} = [U_k \quad U_b \quad U_r], \quad \tilde{V} = [V_k \quad V_b \quad V_r], \quad (17)$$

where

$$U_k \in \mathbb{R}^{m \times k}, \quad U_b \in \mathbb{R}^{m \times p}, \quad V_k \in \mathbb{R}^{n \times k}, \quad V_b \in \mathbb{R}^{n \times p},$$

and

$$\Sigma_k := \text{diag}(\tilde{\sigma}_1, \dots, \tilde{\sigma}_k). \quad (18)$$

The base whitened approximation is

$$M_k := U_k \Sigma_k V_k^\top. \quad (19)$$

Low-dimensional correction parameterization. We restrict the correction to:

1. diagonal updates of the kept singular values;
2. rotations from the buffer into the kept left singular subspace;
3. rotations from the buffer into the kept right singular subspace.

Let

$$D := \text{Diag}(\delta s), \quad P \in \mathbb{R}^{p \times k}, \quad R \in \mathbb{R}^{p \times k}, \quad (20)$$

where $\delta s \in \mathbb{R}^k$. Define the corrected whitened model

$$\widehat{M}(\theta) = (U_k + U_b P)(\Sigma_k + D)(V_k + V_b R)^\top, \quad (21)$$

where

$$\theta := \begin{bmatrix} \delta s \\ \text{vec}(P) \\ \text{vec}(R) \end{bmatrix}.$$

Keeping only first-order terms gives

$$\widehat{M}(\theta) \approx M_k + \delta M(\theta), \quad (22)$$

with

$$\delta M(\theta) = U_k D V_k^\top + U_b P \Sigma_k V_k^\top + U_k \Sigma_k R^\top V_b^\top. \quad (23)$$

Whitened regression target. When compressing layer ℓ sequentially, let X_ℓ^{cur} denote the current calibration inputs produced by the already-compressed prefix, and let

$$Y_\ell^\star := W_\ell X_\ell^{\text{cur}} \quad (24)$$

be the teacher outputs of the original layer on the same inputs. Define

$$X_w := A^{-1/2} X_\ell^{\text{cur}}, \quad Y_w := B^{1/2} Y_\ell^\star. \quad (25)$$

The base residual in whitened coordinates is

$$R_0 := Y_w - M_k X_w. \quad (26)$$

We then solve the ridge-regularized correction problem

$$\min_{\theta} \frac{1}{2} \|R_0 - \delta M(\theta) X_w\|_F^2 + \frac{\lambda}{2} \|\theta\|_2^2. \quad (27)$$

Vectorized closed form. Using $\text{vec}(AXB^\top) = (B \otimes A)\text{vec}(X)$, the first-order correction satisfies

$$\text{vec}(\delta M(\theta)) = J\theta, \quad (28)$$

where

$$J = \begin{bmatrix} V_k \odot U_k & V_k \Sigma_k \otimes U_b & V_b \otimes U_k \Sigma_k \end{bmatrix}. \quad (29)$$

Here $\odot$ denotes the Khatri–Rao product. Since

$$\text{vec}(\delta M(\theta) X_w) = (X_w^\top \otimes I_m) J \theta,$$

define

$$Z := (X_w^\top \otimes I_m) J, \quad r_0 := \text{vec}(R_0). \quad (30)$$

Then (27) becomes

$$\min_{\theta} \frac{1}{2} \|r_0 - Z\theta\|_2^2 + \frac{\lambda}{2} \|\theta\|_2^2, \quad (31)$$

whose exact solution is

$$\theta^\star = (Z^\top Z + \lambda I)^{-1} Z^\top r_0. \quad (32)$$

Buffered-span retraction. Let

$$\bar{U} := [U_k, U_b] \in \mathbb{R}^{m \times (k+p)}, \quad \bar{V} := [V_k, V_b] \in \mathbb{R}^{n \times (k+p)}. \quad (33)$$

From (23), the corrected model inside the buffered span can be written as

$$M_k + \delta M(\theta^*) = \bar{U} C(\theta^*) \bar{V}^\top, \quad (34)$$

where

$$C(\theta^*) = \begin{bmatrix} \Sigma_k + D^* & \Sigma_k (R^*)^\top \\ P^* \Sigma_k & 0 \end{bmatrix} \in \mathbb{R}^{(k+p) \times (k+p)}. \quad (35)$$

Now compute the SVD of the *small* core matrix:

$$C(\theta^*) = U_c \Sigma_c V_c^\top. \quad (36)$$

Retaining the top k singular triplets yields the final corrected whitened rank- k approximation

$$\widehat{M}_k = (\bar{U} U_{c,k}) \Sigma_{c,k} (\bar{V} V_{c,k})^\top. \quad (37)$$

Mapping back to the original parameter space gives

$$\widehat{W}_k = B^{-1/2} \widehat{M}_k A^{-1/2}. \quad (38)$$

The final deployable low-rank factors are

$$W_u = B^{-1/2} (\bar{U} U_{c,k}) \Sigma_{c,k}^{1/2}, \quad W_v = \Sigma_{c,k}^{1/2} (\bar{V} V_{c,k})^\top A^{-1/2}, \quad (39)$$

so that

$$\widehat{W}_k = W_u W_v.$$

1.7 Complexity

The revised method avoids both the dense Hessian and the full tangent system:

1. the base compression uses only $A \in \mathbb{R}^{n \times n}$ and $B \in \mathbb{R}^{m \times m}$, not $H \in \mathbb{R}^{mn \times mn}$;
2. the generalized SVD needs only the top $k + p$ singular triplets of M , which can be computed by randomized SVD;
3. the correction dimension is

$$q = k + 2pk, \quad (40)$$

which is much smaller than $k^2 + k(m - k) + k(n - k)$;

4. the final retraction requires only the SVD of a $(k + p) \times (k + p)$ matrix.

Practical approximation of B . If storing a full $m \times m$ matrix B is too expensive, one may use:

$$B = I_m, \quad B = \text{Diag}(b), \quad \text{or} \quad B \approx \alpha I_m + Q \Lambda Q^\top \quad (41)$$

with a low-rank sketch $Q \Lambda Q^\top$. All three variants preserve the same derivation.

1.8 Reduction to Existing SVD-Based Methods

The proposed method contains simpler SVD-based methods as special cases:

1. If $B = I_m$, the base solution reduces to one-sided whitening.
2. If $B = I_m$ and $p = 0$, the method reduces to plain whitened truncated SVD.
3. If $B = I_m$, $p = 0$, and the calibration objective is $\|(\widehat{W} - W)X\|_F^2$, it recovers the standard activation-whitened low-rank formulation.

[t] Global Rank Allocation by Curvature-Whitened Spectrum [1] For each layer $\ell = 1, \dots, L$: dimensions (m_ℓ, n_ℓ) , generalized singular values $\{\tilde{\sigma}_{\ell,i}\}_{i=1}^{r_\ell}$, target parameter budget P_{budget} . Target ranks $\{k_\ell\}_{\ell=1}^L$.

Initialize $k_\ell \leftarrow r_\ell$ for all ℓ . $P \leftarrow \sum_{\ell=1}^L (m_\ell + n_\ell)r_\ell$. Create an empty list $\mathcal{S}$.

$\ell = 1, \dots, L$ $i = 1, \dots, r_\ell$ Compute score

$$\rho_{\ell,i} \leftarrow \frac{\tilde{\sigma}_{\ell,i}^2}{m_\ell + n_\ell}.$$

Append $(\rho_{\ell,i}, \ell, i)$ to $\mathcal{S}$.

Sort $\mathcal{S}$ in ascending order of $\rho_{\ell,i}$.

each $(\rho, \ell, i) \in \mathcal{S}$ $P \leq P_{\text{budget}}$ **break** $k_\ell \geq i$ $k_\ell \leftarrow k_\ell - 1$ $P \leftarrow P - (m_\ell + n_\ell)$

$\{k_\ell\}_{\ell=1}^L$

[t] Layerwise Bi-Whitened GSVD with Buffered Gauss–Newton Correction [1] Layer weight $W \in \mathbb{R}^{m \times n}$, current calibration inputs $X^{\text{cur}} \in \mathbb{R}^{n \times N}$, output gradients $G \in \mathbb{R}^{m \times N}$ (optional), target rank k , buffer size p , regularization $\varepsilon_A, \varepsilon_B, \lambda$. Compressed factors $W_u \in \mathbb{R}^{m \times k}$, $W_v \in \mathbb{R}^{k \times n}$.

Compute

$$A \leftarrow \frac{1}{N} X^{\text{cur}} (X^{\text{cur}})^\top + \varepsilon_A I_n.$$

output gradients are available Compute

$$B \leftarrow \frac{1}{N} G G^\top + \varepsilon_B I_m.$$

$B \leftarrow I_m$.

Compute whitening factors $A^{1/2}, A^{-1/2}, B^{1/2}, B^{-1/2}$. Form the whitened matrix

$$M \leftarrow B^{1/2} W A^{1/2}.$$

Compute the top $(k + p)$ singular triplets of M :

$$M \approx [U_k \ U_b] \begin{bmatrix} \Sigma_k & 0 \\ 0 & \Sigma_b \end{bmatrix} [V_k \ V_b]^\top.$$

Set base whitened approximation

$$M_k \leftarrow U_k \Sigma_k V_k^\top.$$

Compute teacher outputs on the same current inputs

$$Y^* \leftarrow W X^{\text{cur}}.$$

Whiten the regression problem:

$$X_w \leftarrow A^{-1/2} X^{\text{cur}}, \quad Y_w \leftarrow B^{1/2} Y^*, \quad R_0 \leftarrow Y_w - M_k X_w.$$

Build the correction operator

$$J \leftarrow [V_k \odot U_k \quad V_k \Sigma_k \otimes U_b \quad V_b \otimes U_k \Sigma_k],$$

and

$$Z \leftarrow (X_w^\top \otimes I_m) J, \quad r_0 \leftarrow \text{vec}(R_0).$$

Solve the ridge system

$$\theta^* \leftarrow (Z^\top Z + \lambda I)^{-1} Z^\top r_0.$$

Parse

$$\theta^* \mapsto (\delta s^*, P^*, R^*), \quad D^* \leftarrow \text{Diag}(\delta s^*).$$

Form the small buffered core

$$C \leftarrow \begin{bmatrix} \Sigma_k + D^* & \Sigma_k (R^*)^\top \\ P^* \Sigma_k & 0 \end{bmatrix}.$$

Compute the SVD of C :

$$C = U_c \Sigma_c V_c^\top.$$

Retain the top k singular triplets:

$$U_{c,k} \leftarrow U_c(:, 1:k), \quad \Sigma_{c,k} \leftarrow \Sigma_c(1:k, 1:k), \quad V_{c,k} \leftarrow V_c(:, 1:k).$$

Set

$$\bar{U} \leftarrow [U_k, U_b], \quad \bar{V} \leftarrow [V_k, V_b].$$

Form final compressed factors

$$W_u \leftarrow B^{-1/2} (\bar{U} U_{c,k}) \Sigma_{c,k}^{1/2}, \quad W_v \leftarrow \Sigma_{c,k}^{1/2} (\bar{V} V_{c,k})^\top A^{-1/2}.$$

W_u, W_v

[t] Network-Wide Compression [1] Layers $\{W_\ell\}_{\ell=1}^L$, calibration set $\mathcal{D}$, target model budget P_{budget} , buffer size p , regularization $\varepsilon_A, \varepsilon_B, \lambda$. Compressed model $\{\widehat{W}_\ell\}_{\ell=1}^L$.

Using the original model and calibration set $\mathcal{D}$, estimate per-layer statistics A_ℓ, B_ℓ and generalized singular values $\{\tilde{\sigma}_{\ell,i}\}$. Compute target ranks $\{k_\ell\}_{\ell=1}^L$ with Algorithm 1.8.

$\ell = 1, \dots, L$ Run the already-compressed prefix on $\mathcal{D}$ to obtain current inputs X_ℓ^{cur} . Obtain output gradients G_ℓ if available; otherwise set $B_\ell = I$. Apply Algorithm 1.8 to obtain $W_{u,\ell}, W_{v,\ell}$. Replace W_ℓ by $W_{u,\ell} W_{v,\ell}$.

the compressed network

1.9 Summary

The revised method is practical because:

1. it never forms the dense Hessian H ;
2. the base compression is an exact generalized SVD under a Kronecker curvature model;
3. the post-truncation update is confined to a buffered subspace of size $k + p$;
4. the correction solves a ridge problem with only

$$q = k + 2pk$$

unknowns;

5. the final model remains in low-rank factorized form for efficient inference.

This makes the method substantially more feasible than the original full-Hessian tangent-OBS idea, while remaining strictly stronger than one-sided whitened SVD.

References